Assignment1 **Deadline: 29/06/26**

Study of networking diagnostic & monitoring tools and Socket programming system calls

1a. Apply and analyze the use of various networking diagnostic and monitoring tools in a computer network environment. Demonstrate the execution of the following tools and record your observations regarding network status, configuration, connectivity, routing, and performance:

- **netstat**
- **ifconfig**
- **iwconfig**
- **ping**
- **route**
- **nmap**
- **tcpdump**
- **dig**
- **traceroute**
- **mtr**
- **curl**
- **wget**
- **tracepath**
- **nslookup**
- **host**
- **whois**

1b. Study and analyze the functionality of the following socket programming system calls used in network communication:

- **socket()**
- **bind()**
- **listen()**
- **connect()**
- **accept()**
- **close()**
- **bzero()**
- **htons()**
- **htonl()**
- **ntohs()**
- **ntohl()**

- **read()**
- **write()**
- **send()**
- **recv()**
- **sendto()**
- **recvfrom()**
- **select()**
- **setsockopt()**
- **fcntl()**
- **getpeername()**
- **gethostname()**
- **gethostbyname()**
- **gethostbyaddr()**

Explain the purpose, syntax, parameters, return values, and applications of each function in TCP/IP and UDP socket programming, and record the observations.